

SKILL – 16

Code:

```
GNU nano 7.2 skill16_background_jobs.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

#define MAX_JOBS 10

typedef enum {
    RUNNING,
    DONE
} JobState;

typedef struct {
    int job_id;
    pid_t pid;
    pid_t pgid;
    JobState state;
    int duration;
} Job;

Job job_table[MAX_JOBS];
int job_count = 0;
int next_job_id = 1;

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo     M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_/ Go To Line M-E Redo     M-G Copy
```

Output:

Background process:

```
background-shell> run 5
[Job 1] Started background process PID=548 PGID=548
Prompt returned immediately. Parent is not blocked.

background-shell> Child process PID 548 started.
Child process PID 548 finished.
jobs

[Job 1] PID 548 completed.
```

```
[Job 1] PID 548 completed.

===== JOB TABLE =====
ID      PID      PGID      STATE      DURATION
1        548      548      DONE       5 sec
=====

background-shell> jobs

===== JOB TABLE =====
ID      PID      PGID      STATE      DURATION
1        548      548      DONE       5 sec
=====
```

```
background-shell> cleanup  
Removing completed Job 1 (PID 548)
```

```
background-shell> jobs
```

```
===== JOB TABLE =====  
No active jobs.  
=====
```

```
background-shell> exit
```

```
All background jobs cleaned up.  
Program terminated.
```